# ROHAN RAMESH

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## EDUCATION

**APJ Abdul Kalam Technological University — B.Tech in Computer Science | GPA: 8.3 / 10** 2020 – 2024
**Kerala State Board — Class XII | GPA: 9.3 / 10** 2019 – 2020
**Kerala State Board — Class X | GPA: 9.9 / 10** 2017 – 2018

## TECHNICAL SKILLS

**Programming Languages:** Java, Python, SQL
**Frameworks & Libraries:** Spring Boot, React, Vaadin, JUnit, JPA/Hibernate, Langchain, Langgraph
**Cloud & DevOps:** AWS, Vercel, Render, Docker, Portainer, Linux, Shell Scripting
**Databases:** PostgreSQL, MySQL
**Tools & Technologies:** Git, GitHub, Maven, LLMs, RAG, REST APIs, Postman, Swagger, Jira, Eclipse, JProfiler, Claude Code, New Relic
**Testing:** Unit Testing, Manual Testing

## PROFESSIONAL EXPERIENCE

**Metquay Inc. | Junior Software Developer – Java Full Stack** **Jun 2025 – Present**
*Trivandrum, Kerala*

- Develop and maintain enterprise-level backend applications using Core Java, Spring Framework, and Spring Boot, ensuring scalability and performance.
- Build responsive and dynamic frontend interfaces using React and Vaadin aligned with UX best practices.
- Design, develop, and integrate RESTful APIs for internal and external service communication.
- Manage PostgreSQL databases including schema design, complex query optimization, and data management.
- Deploy and manage applications using AWS and Portainer for cloud-based environments.
- Conduct unit testing and manual testing to ensure code quality, reduce defects, and maintain software reliability.
- Implement secure authentication and authorization mechanisms using Spring Security and JWT-based security practices.
- Integrated Zoho Books for automated accounting and financial data synchronization.
- Collaborate with cross-functional teams in an Agile development environment, participating in sprint planning, code reviews, and feature delivery.
- Optimize application performance through efficient backend logic, database query tuning, and API enhancements.

**Quest Global | Trainee Engineer Intern – Java & Android** **Nov 2024 – Mar 2025**
*Trivandrum, Kerala*

- Developed case-study applications using Core Java principles, demonstrating solid understanding of object-oriented programming.
- Built Android applications using Java and Kotlin, applying mobile development best practices.
- Authored Python automation scripts to streamline repetitive tasks and improve team efficiency.
- Participated in debugging, testing, and software quality improvement activities under senior guidance.
- Worked with Linux command-line utilities for environment setup, file management, and script execution.

## PROJECTS

**Job Application Platform**

- Built a full-stack job portal with features including job posting, job applications, saved jobs, and a real-time chat system.
- Implemented role-based authentication and authorization using Spring Security and JWT, supporting separate Admin and User access.
- Developed RESTful APIs for job management, user management, application tracking, and saved-job functionality.
- Integrated a real-time messaging system using WebSockets for seamless communication between users and administrators.
- Added AI-powered chatbot functionality to provide automated user assistance and improve engagement.
- Implemented file upload and management for resumes and job-related documents.
- Designed and optimized relational database schemas, ensuring data integrity and efficient query performance.
- Built responsive and interactive user interfaces using React with reusable components and state management.
- Tech Stack: Java, Spring Boot, Spring Security, JWT, WebSocket, React, PostgreSQL.

**Underwater Image Restoration and Object Detection**

- Developed a machine-learning-based web application to restore degraded underwater images and detect marine objects with improved accuracy.
- Applied image enhancement and restoration techniques to reduce noise, improve visibility, and correct color distortion in underwater images.
- Integrated object detection models to identify and classify underwater entities from restored images.
- Built a user-friendly web interface for image upload, processing, and result visualization.
- Optimized image-processing workflows to improve detection performance and reduce processing time.
- Managed image storage, metadata handling, and database integration for processed results.
- Tech Stack: Python, Machine Learning, React, JavaScript, MySQL.

**Movie Recommendation System**

- Built a movie recommender (React, FastAPI) using LangChain (LCEL) with Google Gemini for language-, genre-, and year-based suggestions.
- Designed prompts for structured JSON output and implemented fallback parsing to handle inconsistent LLM responses.
- Fixed unreliable LLM-generated IMDb links by verifying each suggested title in parallel against the OMDb API, keeping only real movies.
- Streamed results over newline-delimited JSON so cards appear as each title is verified, with a live progress bar.
- Built an interactive UI with year slider, genre/language chips, "Surprise me", filter and sort, detail view and a saved watchlist.
- Tech Stack: Python, LangChain, Google Gemini, FastAPI, React, Vite, JavaScript, OMDb API.

## CERTIFICATIONS

- Google IT Automation with Python Professional Certificate – Coursera

## ACHIEVEMENTS

- Certificate of Participation – National Level Workshop on Stack Development
- Certificate of Participation – National Level Workshop on Internet of Things (IoT)
- Certificate of Participation – Quiz Competition, Techfest KETCON 2024 Inter-College Competition